

EzzSensors Website User Manual

What Is This Website For?

This website is a helping tool to **analyze water level data from two sensors placed at the inlet and outlet of a sewage pipe**. It is primarily used to **calculate the lag time** between the two readings, which is essential for comparing flow conditions at both ends of the pipe. By aligning readings using the calculated lag time, the tool enables detection of **unexpected flow, leakage, or infiltration** in the pipe system.

It is a **simple website tool** that requires just a few input values and provides **fast, interpretable results** for infiltration assessment.

 **Note:** Some **experimental calibration** or field testing may be necessary to understand the **margin of error in estimated lag time**, especially in real-life variable conditions.

Foreign Water Detection in Sewage Pipes

This method allows you to **detect and measure foreign water infiltration** (such as groundwater or stormwater) into sewage systems using sensor data and mathematical modeling.

Tools & Sensors Needed

1. Level sensors with data loggers (at least 2)
 2. IP65/IP68 box to protect sensors
 3. Inclinator (for pipe slope)
 4. Laser distance meter or tape measure (for pipe dimensions)
 5. Data storage/database
 6. Mathematical processing model
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Idea in Simple Steps

1. Install level sensors at the start and end of the pipe.

2. Measure the slope and diameter using an inclinometer and tape measure (verify at both ends).
 3. Data loggers will record (or transmit) level readings.
 4. Process the raw data before storing it in your database.
 5. Apply the following model to interpret water flow and detect foreign water.
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Mathematical Model Summary

Step 1: Calculate the angle θ from the water level h and known pipe radius r .

Step 2: Use θ to compute:

- **Cross-sectional area (A)** of the water
- **Wetted perimeter (P)**

Step 3: Calculate the **hydraulic radius (Rh)**:

$$Rh = A / P$$

Step 4: Apply **Manning's equation** to estimate flow velocity:

$$V = (1/n) * Rh^{(2/3)} * S^{(1/2)}$$

Where:

- n = Manning's roughness coefficient
- S = pipe slope

Step 5: Calculate:

- **Lag Time** = Pipe Length / Velocity
- **Flowrate (Q)** = Velocity $\times$ Cross-sectional Area

Step 6: Compare inlet flowrate to outlet flowrate (after lag time adjustment) to determine foreign water volume:

$$\text{Foreign Water} = Q_{\text{outlet}} - Q_{\text{inlet}}$$

Step-by-Step Guide (Manual Use)

1. Open the Website

Go to: <https://seif882.github.io/Ezzsensors>

2. Enter First Sensor Reading

- **Time:** Enter the time of the first reading (HH:MM:SS)
- **Inlet Depth:** Enter the water depth from the inlet sensor (in mm)
- **Slope:** Enter the pipe slope (example: 0.01)
- **Pipe Diameter:** Enter the diameter of the pipe (in mm)

3. Click Submit

This opens the second input field.

4. Enter Pipe Length

Once the lag time appears, enter the **Pipe Length** in meters.

5. Click submit Again

Click the second submit button to calculate the lag time.

6. Copy the New Time

The **time** in the field labeled "**Enter h outlet (water depth at pipe outlet)**" will now be **auto-filled**. This is the time you should match with the second sensor.

7. Find the Matching Outlet Depth

Go to your second data logger and look for the water depth at the time shown in step 6.

8. Enter Outlet Depth

Type the matched **Outlet Depth** into the field.

9. Click Final Submit

The website now shows the **final result** at the bottom (green box).

10. Record the Result

You can manually record or copy this result into your spreadsheet for analysis.

Limitations to Keep in Mind

- Manning's equation considers the inner surface roughness of the pipe, not dirt or debris.
 - This method currently calculates infiltration **volume**, not source.
 - The time format must be HH:MM:SS.
 - Only supports one pipe at a time.
 - Pipe diameter, slope, and length must remain constant during the reading period.
 - Website is designed for gravity-flow systems only (not pressurized pipes).
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Things You Should NOT Do

- Don't change the pipe diameter/slope/length during the analysis.
 - Don't paste values in time fields; always type them manually.
 - Don't skip steps or enter data out of order.
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That's It!

Repeat these steps for each reading in your data. If you have many readings, consider using the automated script developed for this website (working on it!).